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Construction and Analysis of Contextual Pun Database in Japanese Using Twitter

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Abstract: There is the impact of the spread of Covid-19 infection, the number of single young people increasing rapidly. In consequence, Japanese society will face an unprecedented increase in elderly single-person household. Using a dialogue system to spread mental health care is an urgent task for future Japanese society. Therefore, improving response accuracy and developing a dialogue system with the same dialogue capability as human beings are essential. For this reason, the demand for humor processing is expected to increase rapidly in the future, but previous studies on humor processing have not focused on the factors affecting how interesting humor is. In this paper, we describe the construction and analysis of a contextual pun database for evaluating the fun of humor with contextual information.

Keywords: dialogue system, humor processing, pun, constructing database.

1 Introduction

Japan currently has problems that include a declining birthrate and an aging population. There is also the impact of the spread of Covid-19 infection, the number of single young people increasing rapidly. In consequence, Japanese society will face an unprecedented increase in elderly single-person household. The most important task to solve this problem is mental health care, such as relieving loneliness and stress. To provide good-quality mental health care, the government must increase the number of home-visit care staff. However, it is difficult to hire the staff due to the rapidly increasing medical costs associated with the aging of society. Therefore, using a dialogue system to spread mental health care is an urgent task for future Japanese society.

Rzepka et al. and Jordan et al. have studied a non-task-oriented dialogue system[1, 2]. These studies obtained some results. However, there were many problems including low response accuracy and lack of excitement in the conversation. In order to solve these problems, it is essential not only to improve response accuracy but also to develop a dialogue system with the same dialogue capability as human beings. On the other hand, there are some differences between human's dialogue and system's dialogue. One of the serious factors is that the systems don't have the ability to recognize and generate humor.

There have been several studies on humor recognition and generation, such as automatic generation of puns[3] and a method to generate answers to riddles as puns[4]. All of these researches had a problem: there was no way to evaluate the fun of the generated humor. Based on this problem, we constructed a large-scale pun database and expanded. However, these studies have not investigated the factors affecting how interesting a pun is such as background knowledge and extra-textual context.

In this paper, we describe the construction and analysis of a contextual pun database for evaluating the fun of humor with contextual information.

2 Contextual Pan Database

2.1 Japanese Pun

The subject of this study is Japanese puns. In Japanese, there are three types of characters as kanji, hiragana and katakana. Kanji is the Ideographic characters. Hiragana and katakana are the phonogram. The katakana is used to express foreign words and onomatopoeia. Puns account for a large percentage of humor in Japanese because of the large number of homonyms in Japanese language.

A pun consists of two words and context within one sentence. The two words are phonetically similar and semantically completely different. These words are defined in two sections: seed expression and transformed expressions. The seed expression is phrase that are the basis of puns. Transformed expression is similar section generated from seed expression in pronunciation.

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Table.1: Cases and percentage in pun-tweets collected

	Cases	Percentage [%]
Total number of pun-tweets	121,566	100.00
Intra-tweet context only	116,249	95.63
Conversational context	5,317	4.37

2.1 Category of Context with Pun

In this study, we collect the pun-tweet, which tweet includes a pun on Twitter[1] for the collection of contextual puns. The context sometimes appears not only a phrase other than a pun in the content but also replies. Therefore, we define two types of the context for the pun-tweet such as intra-tweet context and conversational context. Intra-tweet context is a phrase that represents the background of the tweet and how the pun appeared in the tweet. Conversational context is the content of the tweets before and after the pun-tweet. In our opinion, the fun of a contextual pun depends mainly on the conversation immediately surrounding the pun-tweet. For this reason, we treat the 5 tweets before and after the pun-tweet as conversational context.

2.2 Collection of Contextual Pun

The contextual pun is collected by performing an exact-match search on Twitter using 1,000 puns and cloning the search results. For searching, we use the puns numbered 1 to 1,000 in the pun database[2]. The pun database is a large dataset containing 67,000 puns. Each pun in the pun database is assigned a fun-score which is evaluated on a five-point scale by three people. The evaluation of the fun-score is done by crowdsourcing. For cloning, we use a snscrape[3] which is a scraping tool for social networking service(SNS). It scrapes the Twitter's information such as user profiles, hashtags, threads, and search results. In this work, we collect the content of the pun-tweet and the user reaction such as the number of likes and retweets.

As shown in Table.1, the total number of pun-tweets is 121,566. In these cases, 95.63 percent of the pun-tweet contains intra-tweet context only. In addition, 4.37 percent of the pun-tweet includes the conversational context.

The number of puns included in the contextual pun database is 426 out of the 1,000 puns used to collect for contextual puns. The average cases of pun-tweets per pun is 239.82.

[1] https://twitter.com
[2] http://arakilab.media.eng.hokudai.ac.jp/~araki/dajare.htm
[3] https://github.com/JustAnotherArchivist/snscrape

The percent of pun-tweet with conversational context is tiny: 4.37 percent of total. Therefore, we will collect the pun-tweets using more than 1,001 puns in the pun database in future work. Furthermore, we will also discuss the method for expanding the contextual pun database.

2.3 Detail of Contextual Pun Database

Table.2 shows the tags of the contextual pun database. CPDB-Num is common for the pun-tweet and its conversational context. Reply-Num is 0 for the pun-tweet with the intra-tweet context only and 1-11 for the pun-tweet or its conversational context. Datetime is used to organize the time series of the pun-tweet and its conversational context. Tweet-ID and UserName are used to identify pun-tweets and re-collect the necessary information. Retweets and Likes indicate the number of react to the tweets. In future work, we will analyze the fun of contextual pun using these tags. PTORREP divides the pun-tweet or the conversational context by labeling the pun-tweet with "p" and the conversational context with label "c".

The example of the contents contained in contextual pun dataset is shown Table.3. The number of the tags in Table.3 are the seven without Datetime, Tweet-ID, and UserName.

CPDB-Num 573 has a pun-tweet with conversational context. PTORREP labels indicate that Reply-Num 2 is the pun-tweet. Reply-Num 1 and 3 are the conversational context. The pun-tweet has an advanced conversational context. In Tweet-content of Reply-Num 1, a user tweeted that he/she loves the atmosphere where someone makes a pun, and everyone says "…What?". To this tweet, a different user responded in Reply-Num 2 makes the pun: "Who says a pun?". The user of Reply-Num 1 finally tweeted "…What?" in Reply-Num 3.

Table.2: Tags in contextual pun database

Tags	Descriptions
CPDB-Num	The index of contextual pun database.
Reply-Num	The index of tweets in conversation.
Pun-Matched	The pun contained in contents.
PDB-Num	The index of pun database.
Datetime	The date and time of tweet.
Tweet-ID	The unique number assigned to a tweet.
UserName	The ID of user who tweeted.
Retweets	The number of retweets to tweet.
Likes	The number of likes to tweet.
PTORREP	The label of pun-tweet or conversational context.

Table.3: Example of pun-tweet in contextual pun database

CPDB-Num	Reply-Num	Pun-Matched	PDB-Num	Tweet-content	Retweets	Likes	PTORREP
573	1	ダジャレを言うのは誰じゃ E: Who says a pun?	41	ダジャレ言って みんなが「...え？」 っていう空気好き E*1: I love the atmosphere when someone says a pun and everyone goes “What?”	0	0	c*1
	2			@ayou258192 ダジャレを言うのは誰じゃ E*1: @ayou258192 Who says a pun?	0	0	p*3
	3			@napawa2 ....え? E*1: @napawa2 …What?	0	0	c

Remarks) *1: English translation, *2: the conversational context, *3: the pun-tweet

3 Analysis of Contextual Pun Database

3.1 Bot Tweet

We analyze the percentage of tweets by bot accounts contained in the contextual pun database. A method of analysis is based on the following 3 steps:

1) List bot-candidates that accounts have tweeted the most likely bot: the content contains a pun only or appears more than once in the database.
2) Determine bot or human in Bot-candidates using Botometer[5].
3) Calculate cases and the percentage of tweets by the bots.

Botometer calculates the probability that an account is a bot in the range of 0.00 to 1.00(Bot-Score) based on Twitter usage, network, polarity, tweet content, and followers of account. In this analysis, we regard the account whose Bot-Score is more than 0.80 as a bot.

As a result of analysis, the number of bot-candidates is 3,840. In these cases, 19.51 percent of bot-candidates is determined as bots by Botometer. Furthermore, the number of pun-tweet by bot is 1,499. Hence, 1.23 percent of tweets by bots is included in the contextual pun database.

3.2. Number of Likes and Retweets

Likes and retweets are often given to tweets that are interesting or amusing to Twitter users. We expect that the fun of a contextual pun depends on the conversational context more than the intra-tweet context. Therefore, we analyze the likes and retweets given to the pun-tweets with two types of the context.

Figure.1 indicates that almost likes of pun-tweets with intra-tweet context only are zero. In contrast, about fifty

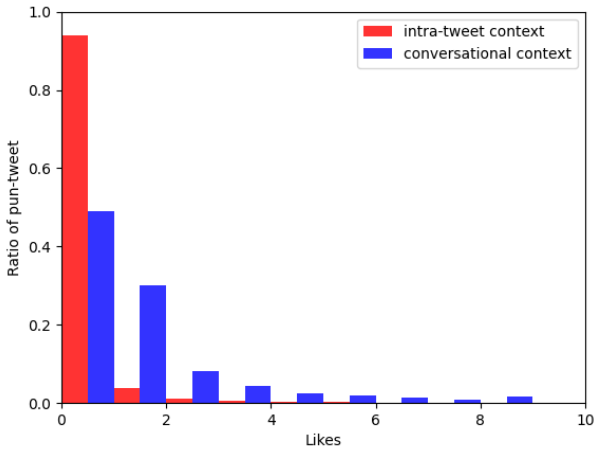


Figure.1: The ratio of pun-tweets with two types of contexts per likes

percent of the pun-tweets with conversational context have more than one likes.

The average of likes given to the pun-tweets with conversational context is 9.40, but the average of likes given to pun-tweets with intra-tweet context is 0.24.

Figure.2 shows that almost retweets of pun-tweet with intra-tweet context only are zero. On the other hand, about ten percent of the pun-tweets with conversational context have more than one retweets.

The average of retweets of the pun-tweets with conversational context is 2.22, but the average of retweets of the pun-tweets with intra-tweet context only is 0.05.

These results show that the number of likes and retweets given to pun-tweets is higher in conversational context than in intra-tweet context. Hence, the conversational context affects to the fun of a contextual pun more than the intra-tweet context.

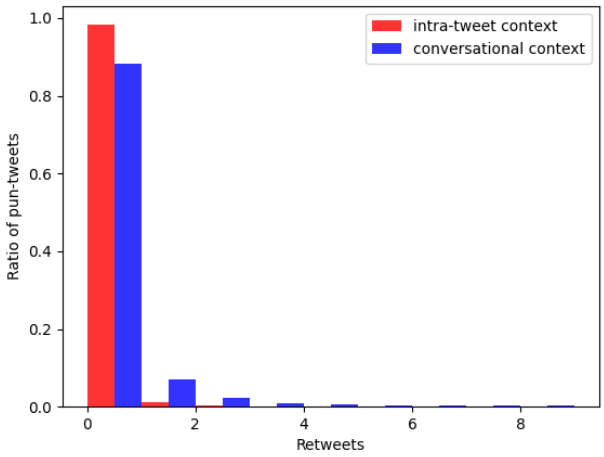


Figure.2: The ratio of pun-tweets with two types of contexts per retweets

3.3. Unrecorded Puns in Contextual Pun Database

The contextual pun database records pun-tweets searched on Twitter using the 1,000 puns in the pun database. However, 574 out of 1,000 puns are unrecorded in the contextual pun database. As the reason for 574 puns unrecord, pun-tweets including these puns are not tweeted by Twitter users. Hence, we analyze the factor that cause 574 puns not to be tweeted by comparing 426 puns recorded.

To analyze this factor, we propose two hypotheses:

(a) Puns unrecorded less interesting than puns recorded.

(b) Puns that include proper nouns are less likely to appear on Twitter.

In order to test these hypotheses, we focus on the fun-scores of puns and the percentage of puns using proper nouns.

3.3.1. Comparing of funnies

Twitter users tweet to get others interested. When tweeting humor on Twitter, users tweet what they find interesting. Hence, as the hypothesis (a), we propose that the fun of puns not to be tweeted lower than the puns recorded in contextual pun database.

To test hypothesis (a), we compare the funniness of 574 puns unrecorded with 426 puns recorded in the contextual pun database using the fun-score of the pun database. The puns in the pun database are given the fun-score by three evaluators. The fun-score is an integer value from 1 to 5 indicating the degree of funnies. Therefore, we use the average of the fun-scores of the three evaluators.

Table.4 indicates that the average of fun-score of the 1,000 puns used to collect the contextual puns is 2.56 points.

Table.4: Average of fun-score

	Cases	Average of fun-score
Puns used to collect	1,000	2.56
Puns unrecorded in contextual pun database	574	2.49
Puns recorded in contextual pun database	426	2.66

The average of fun-score of the puns unrecorded in the contextual pun database is 2.49. On the other hand, the average of fun-score of the puns recorded is 2.66.

Comparing these fun-scores, the average of fun-score of puns unrecorded in contextual pun database is 0.17 points lower than puns recorded. In addition, the average of fun-score of puns unrecorded is 0.07 points lower than puns used in the collection. In contrast, the average of fun-score of puns recorded is 0.11 points higher than puns used in the collection. This result shows that the puns with relatively low fun-scores are not tweeted on Twitter. On the other hand, puns with relatively high fun-scores are tweeted. Therefore, puns unrecorded in contextual pun database less interesting than puns recorded.

3.3.2. Percentage of Puns Using Proper nouns

On Twitter, users from a variety of domains tweet and enjoy conversations with other users. Hence, there are cases where uncommon proper nouns are used in tweets or conversations. However, we consider that humor involving uncommon proper nouns is rarely used on Twitter because only users who know uncommon the proper nouns can enjoy them. Therefore, as the hypothesis (b), we propose that puns including proper nouns are less likely to appear on Twitter.

To test hypothesis (b), we compare the percentage of puns using proper nouns between puns unrecorded and puns recorded. To determine whether a pun contains proper nouns, we use mecab[4] which is a Japanese morphological analyzer. The mecab perform highly accurate analysis and estimation using part-of-speech information.

Table.5 shows that Proper nouns are used in 54.00 percent of the puns unrecorded in the contextual pun dataset. In contrast, 46.71 percent of pans recorded in contextual pun database are used proper nouns.

Comparing the percentage of puns using proper nouns, the puns unrecorded in contextual pun database are 7.29 percent higher than the puns recorded. This result shows that puns unrecorded in contextual pun database has more puns using proper nouns than puns recorded. Hence, puns

[4] https://taku910.github.io/mecab/

Table.5: Cases and percentage of puns using proper nouns

	Cases / Total	Percentage [%]
Puns unrecorded in contextual pun database	310 / 574	54.00
Puns recorded in contextual pun database	199 / 426	46.71

using proper nouns are less likely to appear on Twitter.

4 Conclusion

The demand for humor processing is expected to increase rapidly in the future, but previous studies on humor processing have not focused on the factors affecting how interesting humor is.

For evaluating the fun of humor with contextual information, we construct and analyze a contextual pun database. First, we define two types of the context for the pun-tweet such as intra-tweet context and conversational context. Second, we collect a pun-tweet and user reaction for constructing the contextual pun database. The total number of contextual puns is 121,566. In these cases, 95.63 percent of the pun-tweet contains intra-tweet context only. In addition, 4.37 percent of the pun-tweet includes the conversational context. Third, we analyze the percentage of tweets by bot accounts contained the contextual pun database. As this result, 1.23 percent of tweets by bots is included in the contextual pun database. Fourth, we investigate the likes and retweets given to the pun-tweets with two types of the context. The average of likes given to the pun-tweets with conversational context is 9.40, but the average of likes given to pun-tweets with intra-tweet context is 0.24. The average of retweets of the pun-tweets with conversational context is 2.22 but the average of retweets of pun-tweets with intra-tweet context is 0.05. As these results, the number of likes and retweets given to pun-tweets are higher in conversational context than in intra-tweet context. Hence, the conversational context affects to the fun of a contextual pun more than the intra-tweet context. Fifth, we analyze the factor that cause 574 puns not to be tweeted by comparing 426 puns recorded in contextual pun database. Comparing these fun-scores, the average of fun-score of puns unrecorded in contextual pun database is 0.17 points lower than puns recorded. Therefore, puns unrecorded in contextual pun database less interesting than puns recorded. Comparing the percentage of puns using proper nouns, the puns unrecorded in contextual pun database are 7.29 percent higher than the puns recorded. Hence, puns using proper nouns are less likely to appear on Twitter.

In future work, we will collect the pun-tweets using more than 1,001 puns in the pun database. Furthermore, we will also discuss the method for expanding the contextual pun database.

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REFERENCES

[1] Rafal Rzepka, Shinsuke Higuchi, Michal Ptaszynski, Pawel Dybala and Kenji Araki: When Your Users Are Not Serious, Journal of the Japanese Society for Artificial Intelligence, Vol. 25, No. 1, pp.114-121,2010.

[2] Arnaud JORDAN, Kenji ARAKI: Real-time Language-Independent Algorithm for Dialogue Agents, Journal of Japan Society for Fuzzy Theory and Intelligent Informatics, Vol.28, No.1, pp.535-555, 2016.

[3] Jonas Sjobergh and Kenji Araki: Robots Make Things Funnier, New Frontiers in Artificial Intelligence, Lecture Notes in Artificial Intelligence, Vol.5447, pp.306-313, 2009.

[4] Kim Binsted, Osamu Takizawa: BOKE-A Japanese Punning Riddle Generator, Journal of Japanese Society for Artificial Intelligence, Vol.13, No.6, pp.920-927, 1998.

[5] Mohsen Sayyadiharikandeh, Onur Varol, Kai-Cheng Yang, Alessandro Flammini, Filippo Menczer, Detection of Novel Social Bots by Ensembles of Specialized Classifiers, Proceedings of the 29th ACM International Conference on Information & Knowledge Management, pp. 2725-2732, 2020.